

WEST VALLEY FRESH DISTRIBUTION COMPANY

A disciplined 5-city pilot is the right way to enter organic Hass avocados

This deck argues for a selective launch, not a broad rollout, by testing price support, organic readiness, freight resilience, and seasonality in the supplied city data.

PILOT ORDER
20,000 organic units

MARKET SCOPE
40 city markets

HISTORY
2019 to Jun 2025

Prepared as an anonymized submission for the 2026 IMA Student Case Competition. The deck is designed to stand alone without speaker notes.

DECISION TO PROVE

Win the right five cities first, then earn the right to scale.

THIS DECK TESTS

- Price support beats raw market size.
- The top cities hold through freight shocks.
- Seasonality changes how the pilot should run.

Launch a disciplined 5-city organic pilot

RECOMMENDATION

Launch Seattle, Boise, Portland, Spokane, and San Diego first.

These five cities combine premium June pricing, organic readiness, and manageable freight better than a broad national rollout.

SEATTLE

BOISE

PORTLAND

SPOKANE

SAN DIEGO

AVG PROFIT

\$13.7k

AVG MARGIN

69.7%

RANK-5 GAP

\$1k

WHY THESE CITIES

Premium price markets

The top five also rank as the strongest June organic price markets in the model.

WHY THE LIST HOLDS

Same 5 cities

The city set does not change in the base, +50%, or -50% freight cases.

HOW TO OPERATE IT

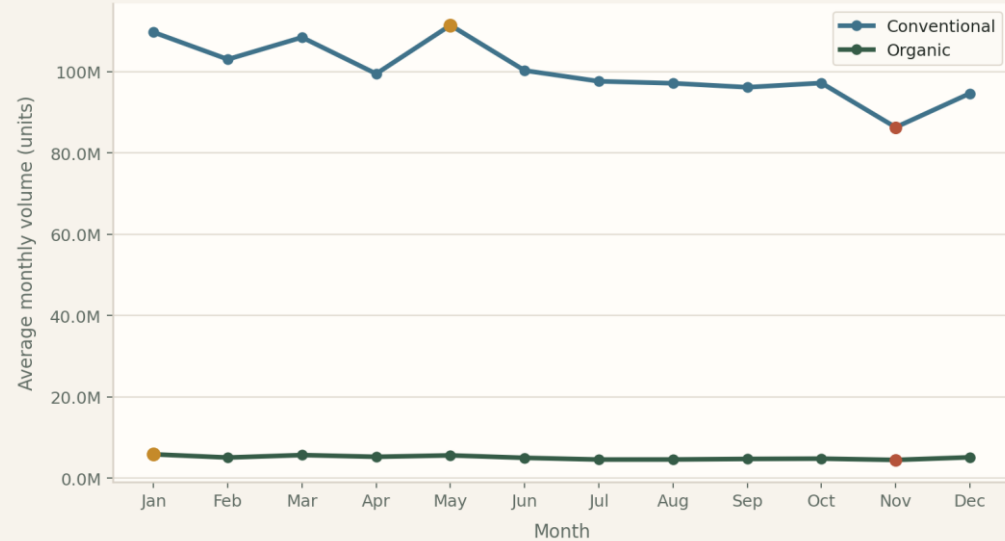
Pilot before scale

Launch narrowly, learn fast, then scale only where margin and service hold.

Model note: each city is scored on the same 20,000-unit scenario using June 2025 price as the June 2026 proxy.

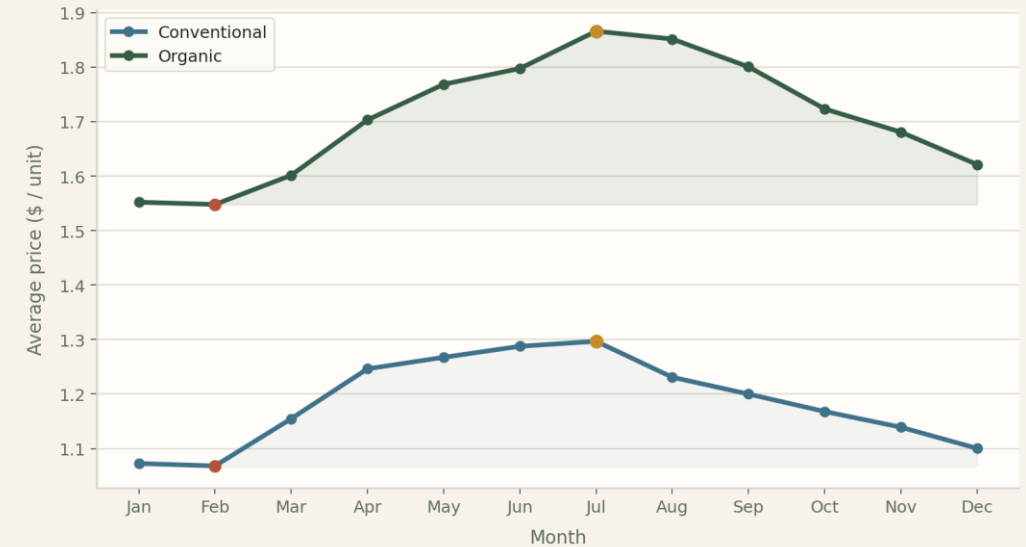
Volume peaks before price

Volume peaks earlier than price, creating a seasonal timing gap



Volume arrives early. Organic volume peaks in January, conventional peaks in May, and both soften into November.

Prices crest in July after the strongest volume months



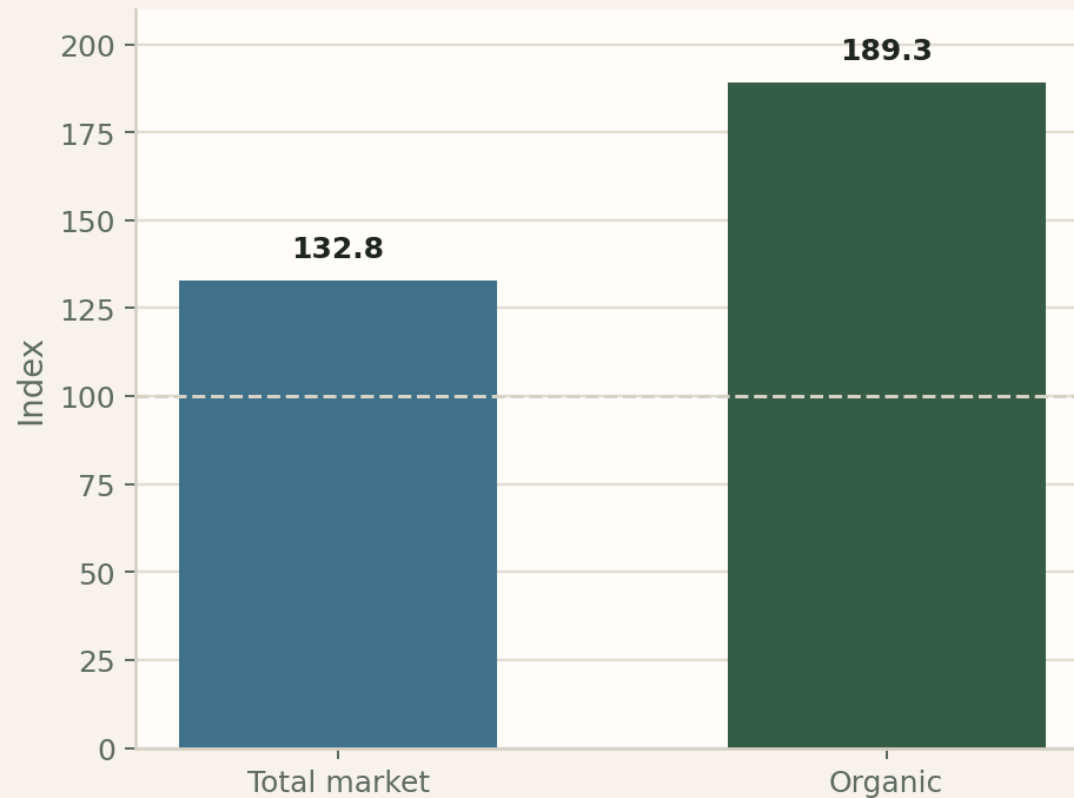
Price crests later. Both organic and conventional prices peak in July and bottom in February.

The market absorbs more units before it offers peak pricing, so timing affects margin capture, promotions, and inventory allocation.

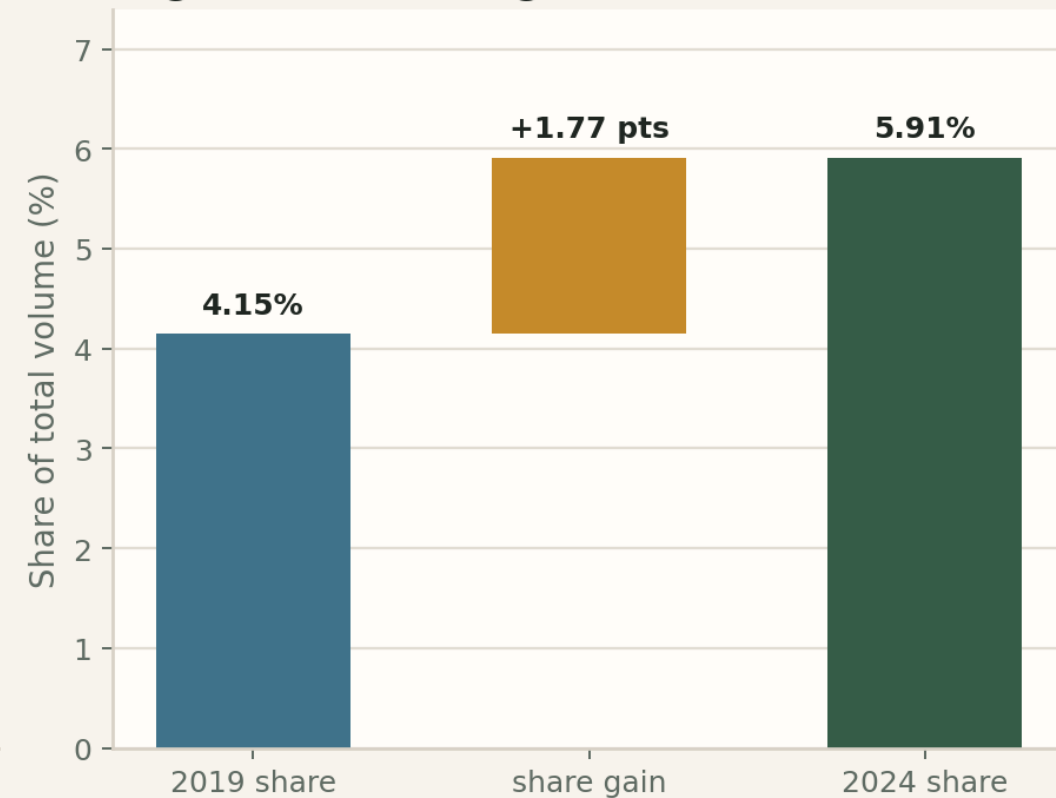
Organic volume nearly doubled from 2019 to 2024 while share rose 1.77 points

Organic volume growth outpaced total market growth by a wide margin.

Indexed volume growth 2019 = 100



Organic share bridge



TAKEAWAY

Organic is still a niche share of the total market, but it is growing materially faster than the overall Hass category.

Use this as market context, not as a launch recommendation by itself.

Three filters should decide the pilot

FILTER	WHAT GOOD LOOKS LIKE	EVIDENCE IN THE MODEL	WHY IT MATTERS
Price support	\$2.09 to \$2.66 June pricing	The selected five are also the five highest-price June 2025 organic markets.	Price support is the first screen, not a nice-to-have.
Organic readiness	Seattle leads 5 of 7 periods	Penetration differs sharply by city, so the launch must favor markets already buying premium mix.	Scale alone is not enough to win an organic pilot.
Route efficiency	\$813 to \$1.0k profit loss at +50% freight	The recommended cities keep the same ranking even after a large shipping shock.	Freight can cut margin, but it does not break the top five.

These three filters explain why the Pacific Northwest plus San Diego survives every screen that follows.

Los Angeles wins Q1 in every period and both types

WINNER HEATMAP

CONVENTIONAL

ORGANIC

2019	LA	\$168.4M	LA	\$9.1M
2020	LA	\$183.5M	LA	\$11.9M
2021	LA	\$185.8M	LA	\$12.7M
2022	LA	\$200.9M	LA	\$11.8M
2023	LA	\$183.9M	LA	\$11.2M
2024	LA	\$211.2M	LA	\$15.3M
2025 H1	LA	\$113.2M	LA	\$8.1M

TAKEAWAY

Q1 is unambiguous: Los Angeles wins both product types in every required period.

WINNER
CELLS
14/14

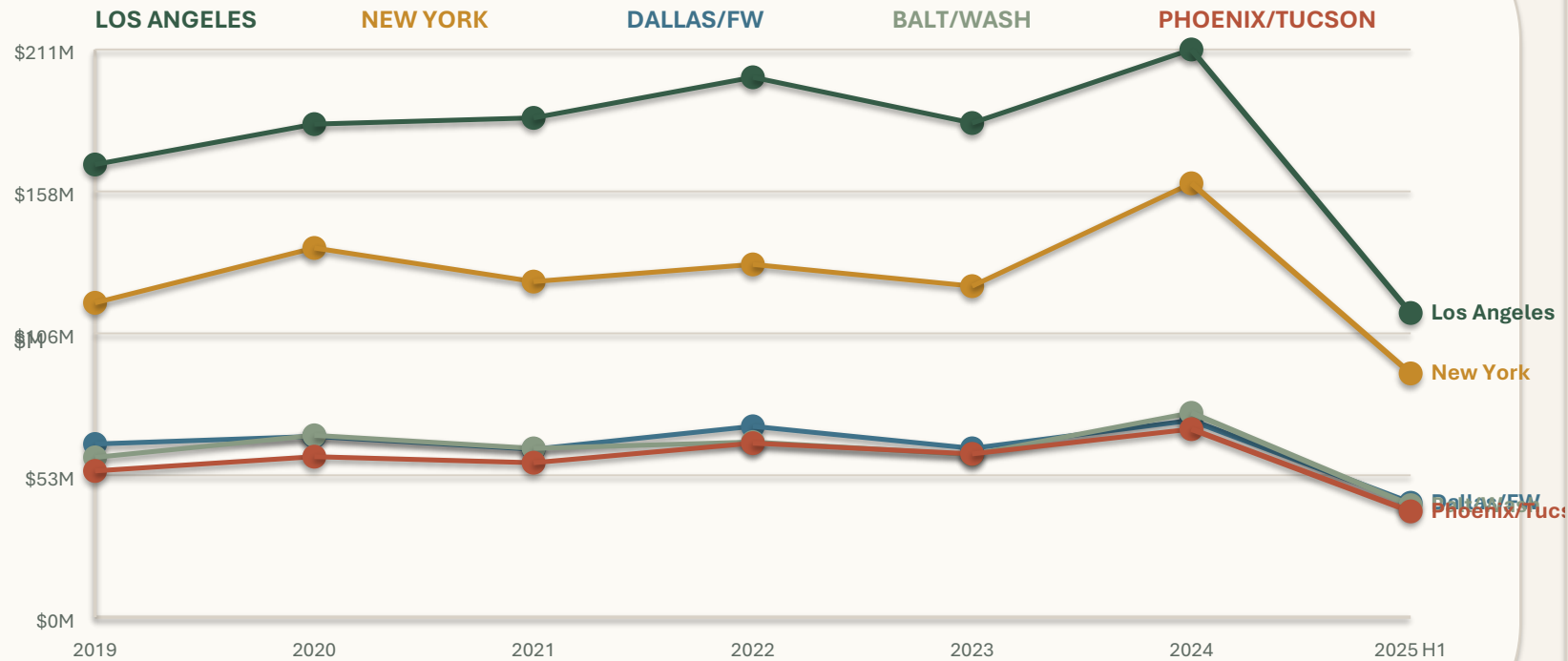
2024 CONV.
\$211.2
M

2024 ORGANIC
\$15.3M

- Los Angeles wins scale decisively.
- That answers Q1 cleanly.
- It does not win the pilot by itself.

Los Angeles is the conventional revenue outlier

TOP 5 CONVENTIONAL REVENUE TREND



2024 LEAD VS NY

\$49.8M

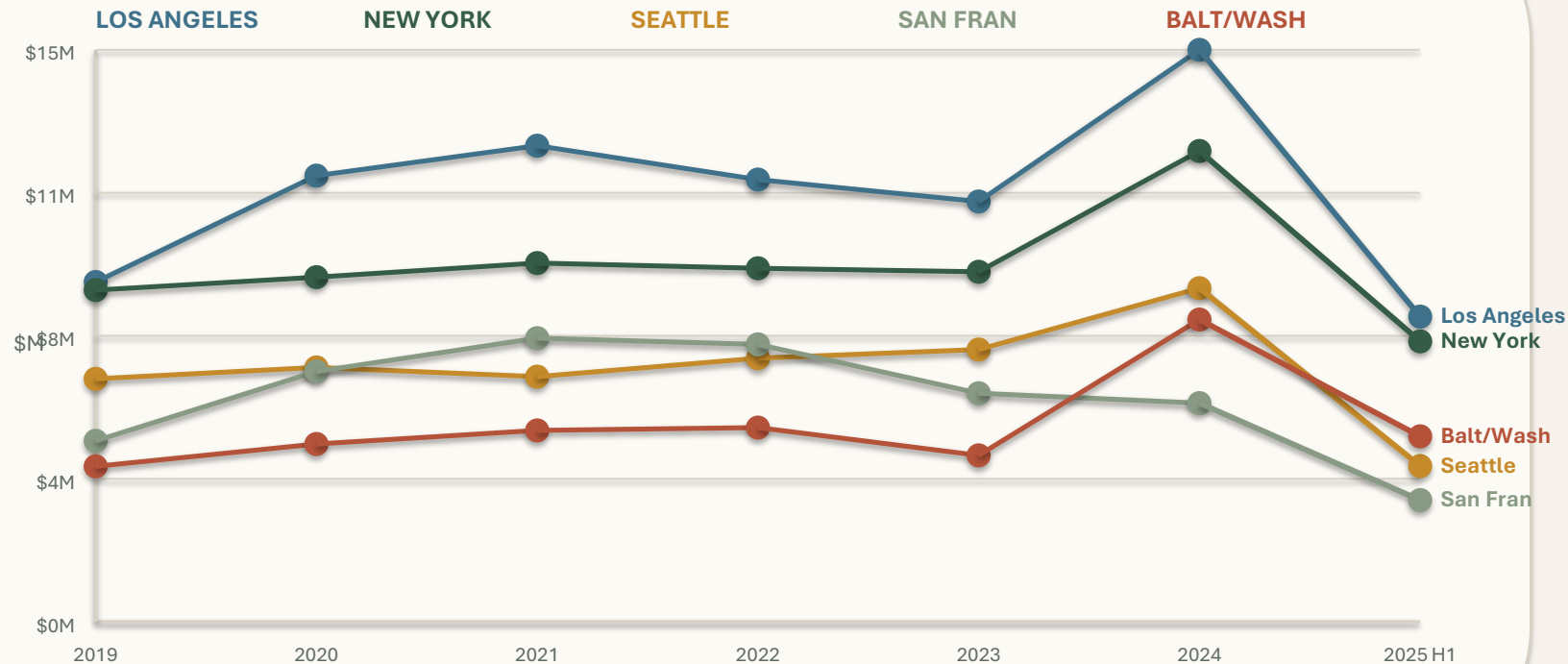
2025 H1 LEAD VS NY

\$22.5M

Los Angeles does not just edge out the field. It runs a different revenue tier.

Los Angeles leads organic sales, but not by much

TOP 5 ORGANIC REVENUE TREND



2024 LEAD VS NY

\$2.7M

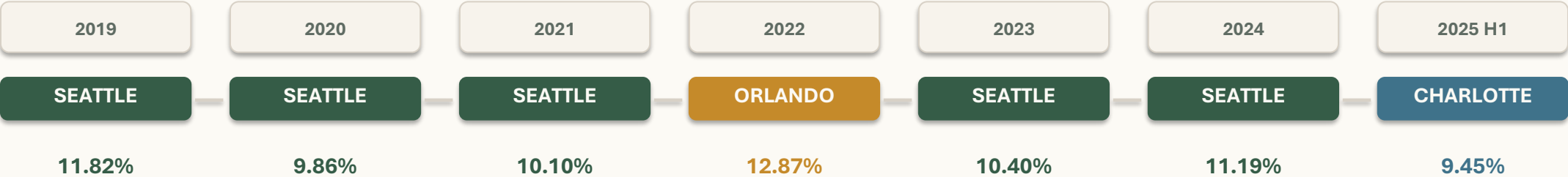
SEATTLE IN TOP TIER

Top 5

Organic is less concentrated than conventional, which is exactly why pilot choice can diverge from raw size.

Seattle leads organic penetration in most periods

WINNER TIMELINE



MOST FREQUENT LEADER

Seattle

Wins 5 of the 7 measured periods and is the clearest indicator of organic readiness.

BEST ONE-YEAR SHARE

Orlando 12.87%

Orlando peaks in 2022, but it does not sustain the same multi-period lead.

CURRENT H1 LEADER

Charlotte 9.45%

Charlotte leads 2025 H1, showing that penetration leadership is not static.

Organic is growing fast but still niche

2019 ORGANIC SHARE

4.15%

Organic started from a small base inside the total Hass market.

2024 ORGANIC SHARE

5.91%

The share has risen, but it is still a minority slice of total demand.

ORGANIC VOLUME GROWTH

+89.3%

Organic volume grew much faster than the total category from 2019 to 2024.

TOTAL VOLUME GROWTH

+32.8%

The overall market also expanded, but at a much slower pace.

WHAT THIS MEANS

- Growth is real enough to justify entry.
- Scale is still small enough that city choice matters.
- That favors a pilot, not a broad launch.

INTERPRETATION

Organic is growing fast, but it is not mature enough to reward a broad, undifferentiated rollout. West Valley should enter selectively.

Higher prices usually mean lower volume

MARKET-LEVEL RELATIONSHIP

PEARSON R

-0.427

R²

0.183

ELASTICITY

-4.96

- Price matters, but not cleanly enough to drive the whole decision.
- Higher-price markets often sell fewer units.
- City choice still needs local market context.

READ IT THIS WAY

Price matters at the market level.

But it does not tell the whole story. Market mix, organic readiness, and local structure still explain why some cities are better pilot markets than others.

That is why the deck separates scale, readiness, and economics instead of forcing one metric to do every job.

Within each type, price explains little

CONVENTIONAL

PEARSON R

-0.026

R2

0.001

ELASTICITY

-0.39

Within each type, price explains little on its own. Local market structure and mix still do the real work.

ORGANIC

PEARSON R

-0.044

R2

0.002

ELASTICITY

-0.81

Within each type, price explains little on its own. Local market structure and mix still do the real work.

Implication: elasticity is context, not the recommendation. City-level economics and readiness still decide the launch.

The launch still faces freight, fuel, labor, and import risk

DRIVER

Freight + fuel

Affects landed cost and changes the margin penalty on long-haul markets.

DRIVER

Imported supply risk

Mexico and cross-border flow disruptions can tighten availability and raise uncertainty.

DRIVER

Labor + inflation

Raises operating pressure throughout sourcing, distribution, and retail execution.

DRIVER

Seasonality + promotions

Changes when the market wants volume and when it supports price.

These drivers raise the cost of being wrong. That is why West Valley should launch narrowly and scale with proof.

LA wins on scale; Seattle and the Northwest win on fit

LOS ANGELES DEMAND FACTORS (INFERRED)

FACTOR

Culinary fit

Avocados are deeply embedded in Mexican and California food culture.

FACTOR

Freshness + familiarity

Proximity to California and Mexican supply likely supports habitual purchase and product confidence.

FACTOR

Retail + foodservice scale

Los Angeles has a massive grocery, restaurant, and foodservice market.

FACTOR

Health-forward demand

Wellness-oriented and higher-income submarkets help support premium produce demand.

SO WHAT FOR THE PILOT

LA ORGANIC SHARE

4.9%

LA JUNE PRICE

\$1.87

SEATTLE SHARE

11.2%

SELECTED-5 AVG

\$2.45

- Los Angeles explains where scale sits.
- Seattle and the Pacific Northwest explain where premium organic entry works better.
- These are inferred demand drivers, not direct causal proof.

Winning entry markets need price, readiness, and manageable miles

ENTRY SCREEN LEADERBOARD

City	Price	Organic share	Miles	Profit	Pilot
Seattle	\$2.66	11.2%	1,029	\$15,266	Selected
Boise	\$2.59	5.4%	827	\$14,838	Selected
Portland	\$2.54	5.9%	855	\$14,408	Selected
Spokane	\$2.38	6.1%	1,097	\$12,992	Selected
San Diego	\$2.09	4.3%	253	\$11,126	Selected
San Francisco	\$1.97	7.0%	283	\$10,118	Not selected
Sacramento	\$1.90	3.7%	277	\$9,578	Not selected
Los Angeles	\$1.87	4.9%	111	\$9,404	Not selected

- Price power filters out large but lower-value markets.
- Organic readiness helps separate premium-fit cities from simple scale leaders.
- Miles still matter, but they do not overturn a strong local price advantage.

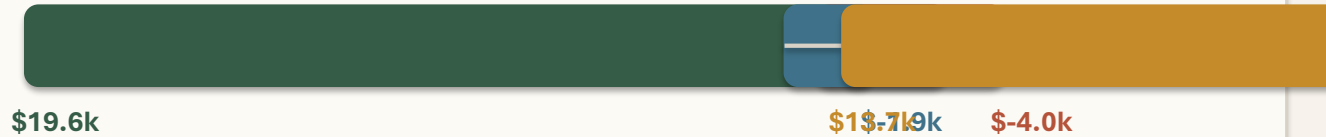
BRIDGE TO RECOMMENDATION

**Best pilot set =
top 5**

The profit model is simple enough to audit

AVERAGE UNIT ECONOMICS

REVENUE



Each city is scored using the same 20,000-unit assumption: planned wholesale revenue minus acquisition cost and the freight formula. The simplicity is a feature because judges can audit the math quickly.

UNITS

20,000

Comparable scorecard for each city.

WHOLESALE PRICE

40% of retail

Revenue proxy used consistently across markets.

FREIGHT

\$1,500

Makes distance visible without overcomplicating the model.

This is not a full allocation model yet. It is a transparent city-selection model.

June pricing makes the Northwest the premium play

TOP JUNE 2025 ORGANIC RETAIL PRICES

Seattle		\$2.66
Boise		\$2.59
Portland		\$2.54
Spokane		\$2.38
San Diego		\$2.09
San Francisco		\$1.97
CincinnatiDayton		\$1.92
Sacramento		\$1.90
Los Angeles		\$1.87
Atlanta		\$1.82

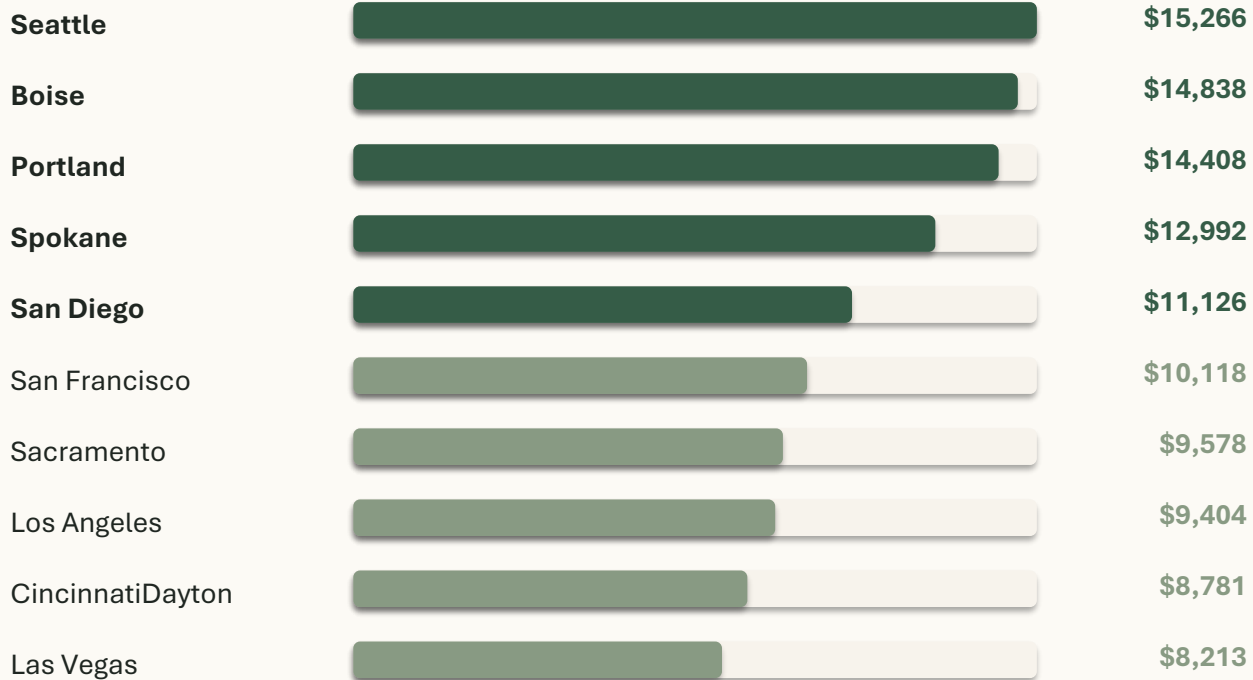
SELECTED CITIES IN
TOP 5
5/5

TOP PRICE
\$2.66

- All five recommended cities also sit in the top five on June price.
- That is why price support keeps beating freight pressure later in the deck.
- Price is the revenue-side reason this pilot wins.

Base-case profit shows a real top-five cut line

TOP 10 CITY PROFIT RANKING



WHY THE CUT LINE IS DEFENSIBLE

TOP-5 AVG

\$13.7k

NEXT-5 AVG

\$9.2k

RANK-5 VS RANK-6 GAP

\$1k

- There is a real cut line after rank five.
- All five win on economics, not just proximity.
- That makes the recommendation easier to defend in the room.

The recommendation is a West Coast cluster

SEATTLE

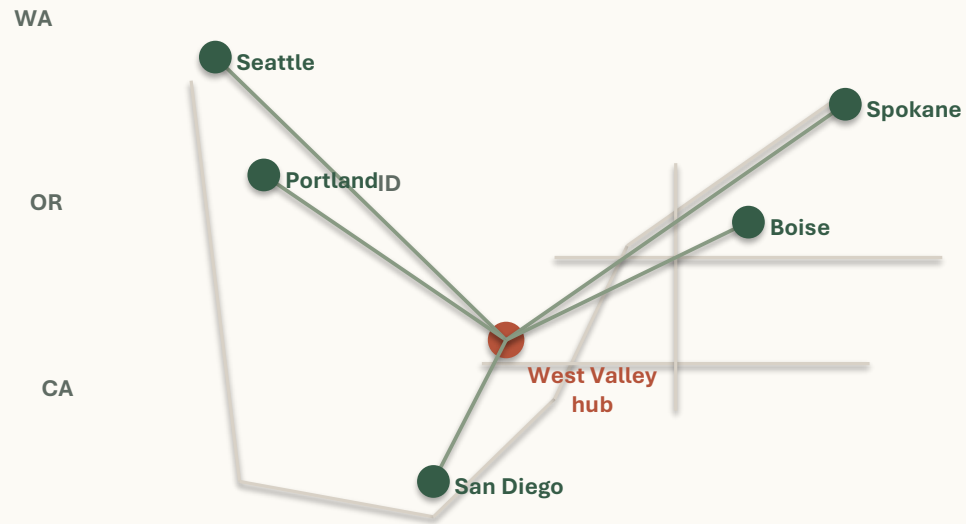
BOISE

PORTLAND

SPOKANE

SAN DIEGO

GEOGRAPHIC CLUSTER



WHY THE CLUSTER WINS

AVG MILES

812

AVG PRICE

\$2.45

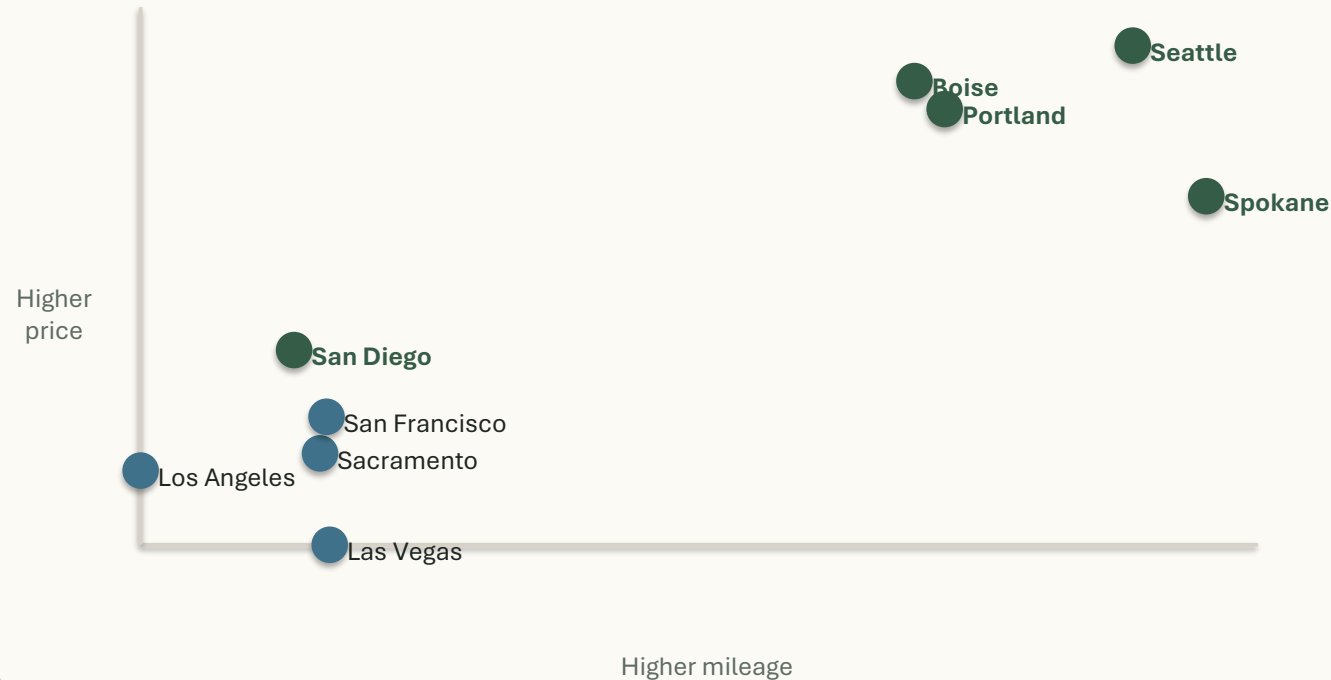
LOW-MILE HEDGE

San Diego

- This is one coherent western lane system, not five disconnected bets.
- The Northwest supplies the premium; San Diego supplies the short-haul hedge.
- That makes the recommendation operational, not just analytical.

Price support beats proximity

PRICE VS MILES



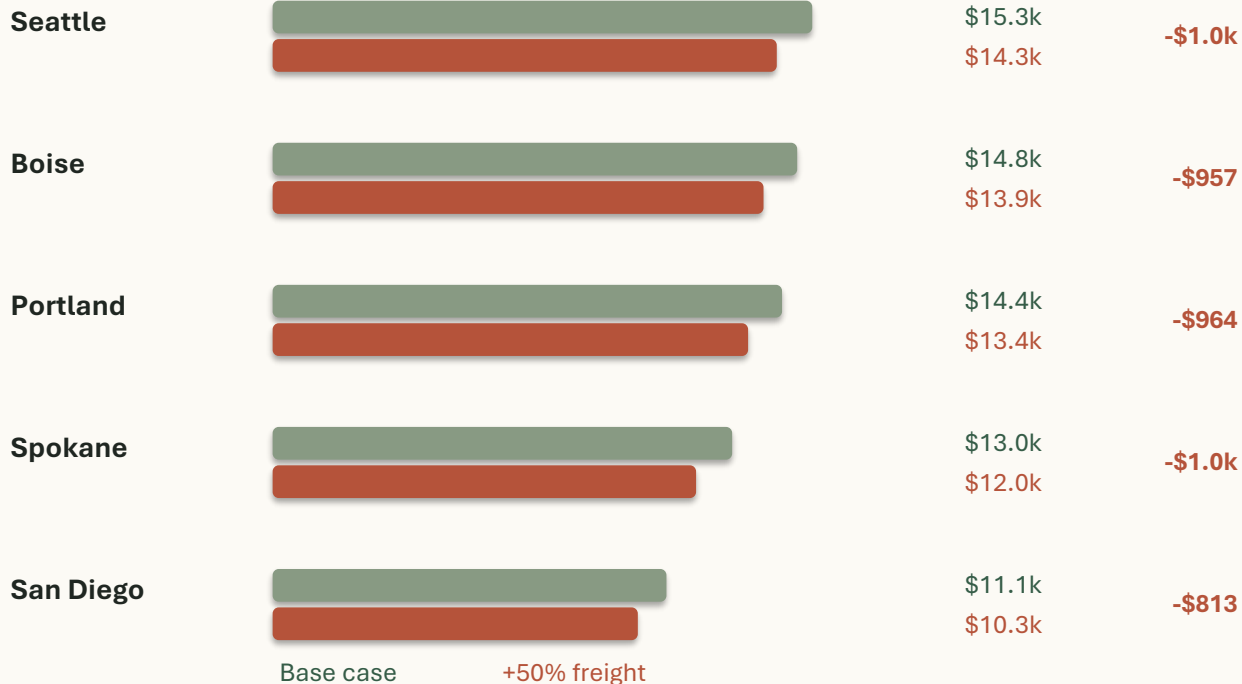
SELECTED AVG
PRICE
\$2.45

NEARBY ALT.
\$1.87

- The winners are not simply the closest cities.
- Nearby alternatives give back the freight edge through weaker pricing.
- San Diego is the hedge that keeps the cluster grounded.

+50% freight cuts margin, not the city set

BASE VS +50% FREIGHT



TOP-5 AVG PROFIT

\$12.8k

AVG MARGIN

64.9%

MARGIN DROP

-4.9 pts

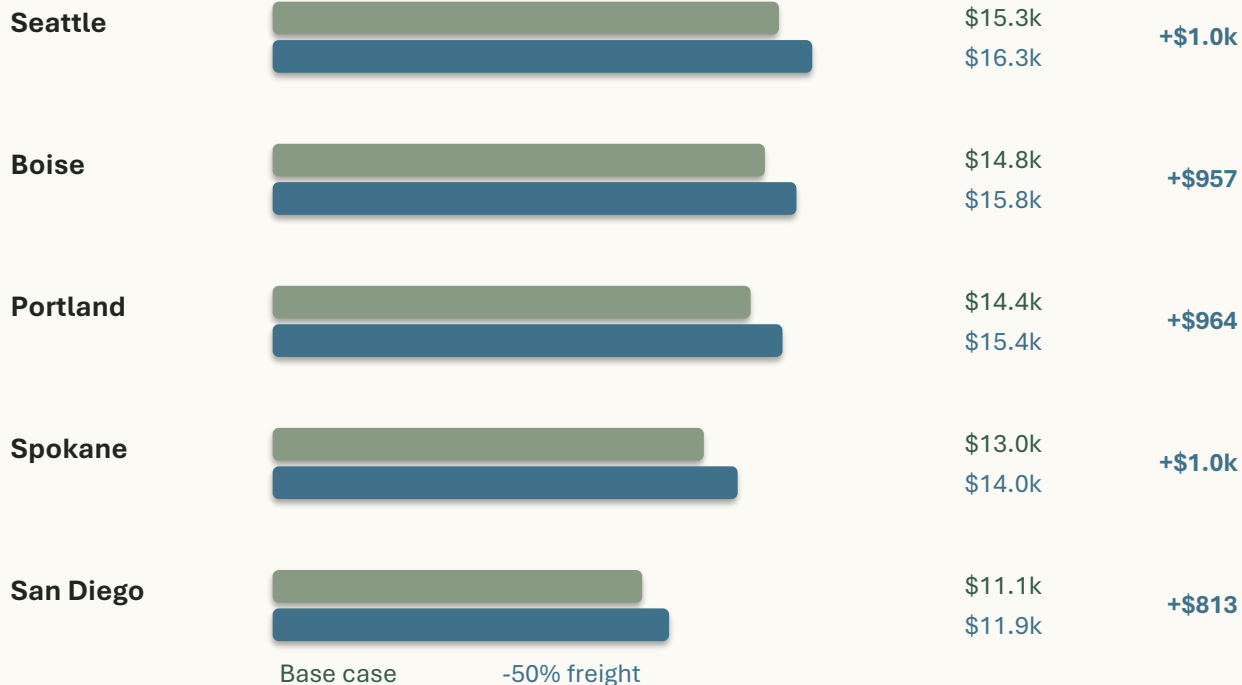
RESULT

Same 5

- This is resilience, not immunity: average margin drops from 69.7% to 64.9%.
- West Valley should hedge fuel exposure and lock carrier capacity early.
- The pilot still holds because price support outruns the freight shock.

-50% freight lifts margin, not the city set

BASE VS -50% FREIGHT



TOP-5 AVG PROFIT

\$14.7k

AVG MARGIN

74.6%

MARGIN GAIN

+4.9 pts

RESULT

Same 5

- Lower freight lifts average margin from 69.7% to 74.6%.
- Use that upside to fund service and trade support, not to chase more cities.
- The same five still win, which confirms this set is strong, not fragile.

The same five cities win in every freight case

RANK MATRIX

Rank	Base	+50%	-50%
1	Seattle	Seattle	Seattle
2	Boise	Boise	Boise
3	Portland	Portland	Portland
4	Spokane	Spokane	Spokane
5	San Diego	San Diego	San Diego

RANK CHANGES

0

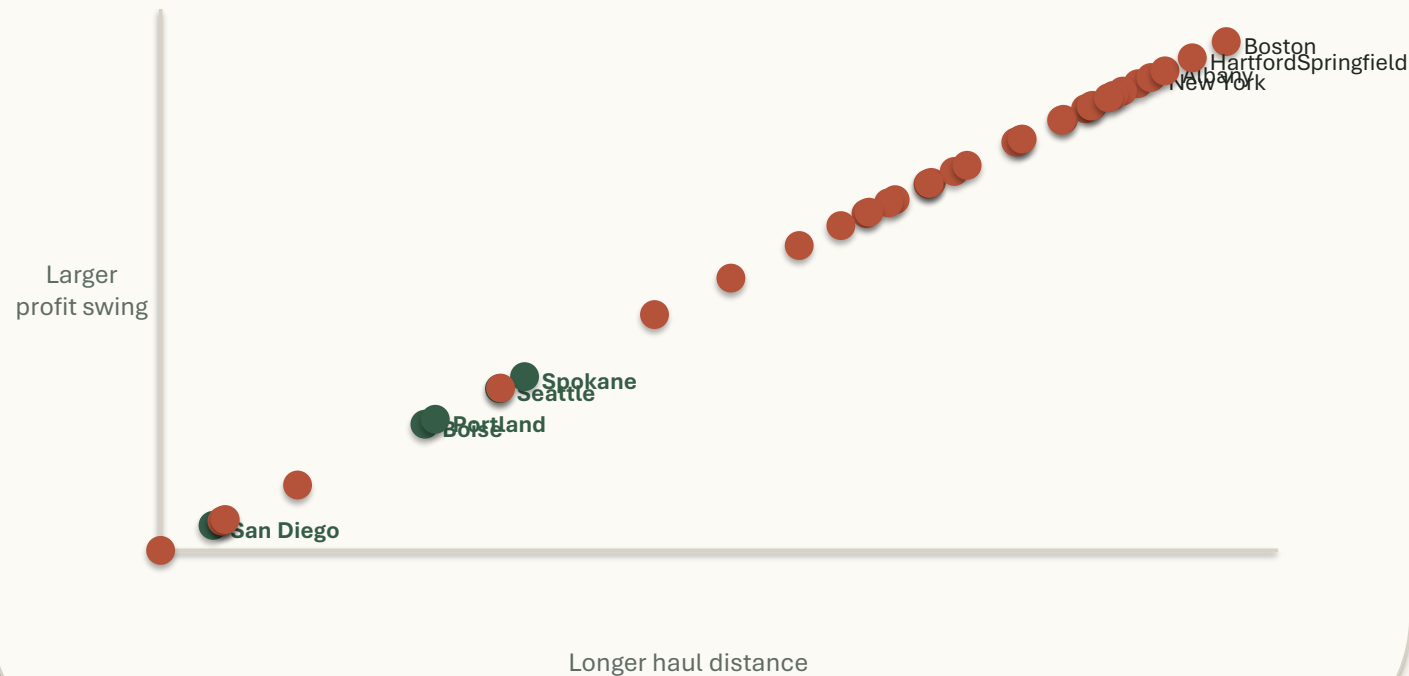
SELECTED SET

Same 5 cities

- Base, upside, and downside freight all return the same five cities in the same order.
- That is the cleanest proof that the recommendation is structurally stable.
- The next slide shows why: price support wins the tie-break.

Distance drives freight risk

MILES VS FREIGHT EXPOSURE



HOW TO READ IT

- The slope is the point: longer-haul cities take the biggest hit.
- The recommended cities are materially less exposed than the Northeast and Midwest tail.
- That makes a western pilot the smarter first move for a new entrant.

SELECTED AVG

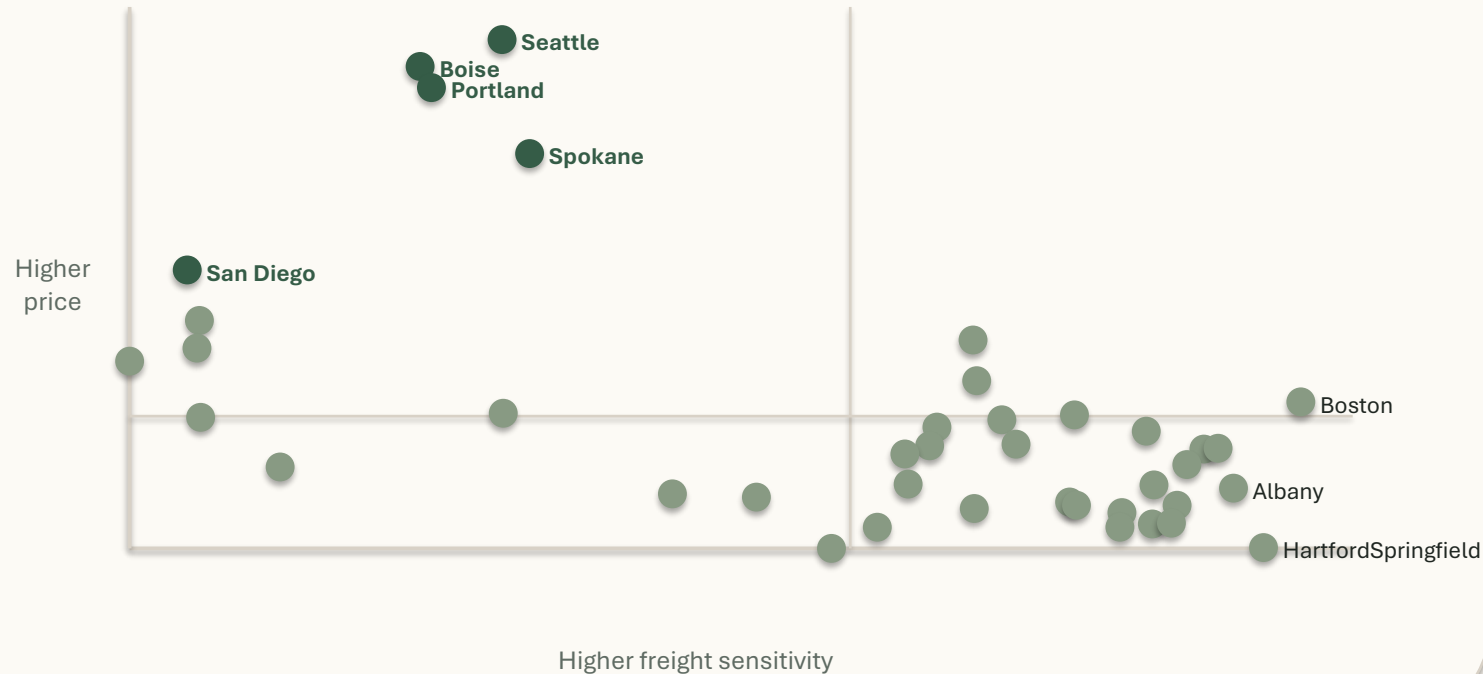
\$953

EXPOSED AVG

\$1,464

Price support decides the set

PRICE SUPPORT VS FREIGHT EXPOSURE



INTERPRETATION

SELECTED AVG
PRICE
\$2.45

SELECTED AVG
SWING
\$953

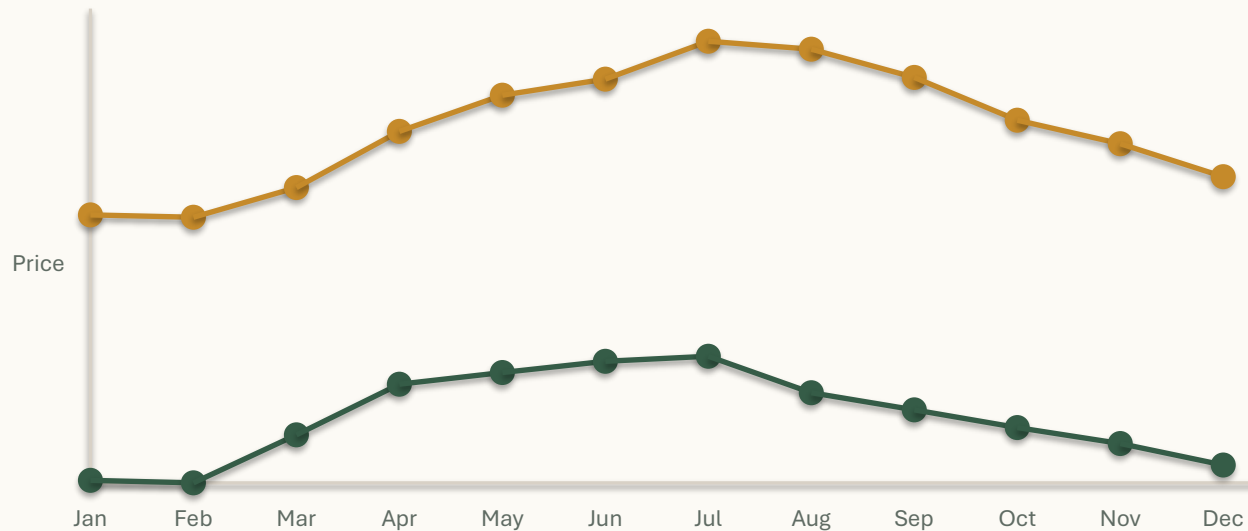
- The recommended cities sit in the premium-price half of the chart, not the freight-risk tail.
- That is why the ranking holds even when margins compress.
- Price support chooses the cities; freight discipline protects the margin.

Prices reach their high point in July

MONTHLY PRICE PROFILE

CONVENTIONAL

ORGANIC



CONVENTIONAL PEAK

Jul

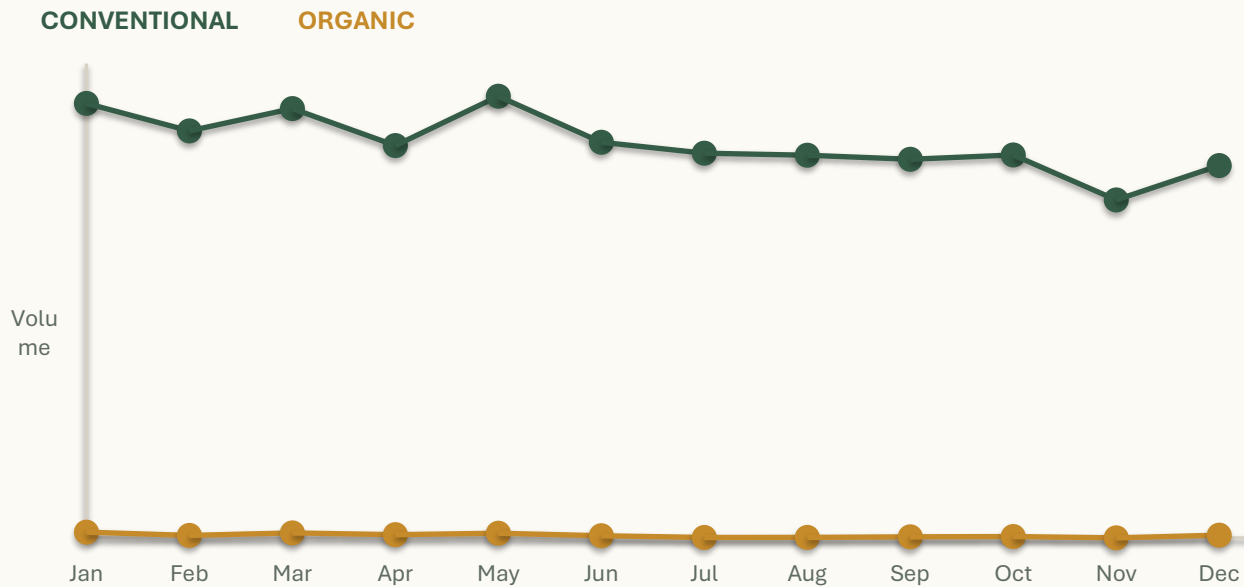
ORGANIC PEAK

Jul

- Both product types peak in July, not during the highest-volume months.
- That means the operating plan should switch from service protection to margin protection in summer.
- Summer is the cleanest season for monetizing premium markets.

Volume peaks earlier than price

MONTHLY VOLUME PROFILE



CONVENTIONAL PEAK

May

ORGANIC PEAK

Jan

- Organic volume peaks in January while conventional peaks in May.
- Both soften into November, which is why off-peak demand support matters.
- Volume timing and price timing do not line up, creating the duck-curve operating problem.

Seasonality should change when West Valley pushes volume vs margin

Organic volume peaks in Jan while price peaks in Jul; that gap is the operating lever.

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

OPERATING STANCE

WINDOW

WHAT WEST VALLEY SHOULD DO

WHY IT MATTERS

Push volume**Jan to May**

Prioritize service, retailer availability, and clean flow while the market is naturally absorbing more units.

This is the best window to prove service quality before summer pricing peaks.

Protect margin**Jul to Sep**

Hold pricing discipline and allocate the strongest fruit to the highest-paying pilot markets.

Price support is strongest here, so margin capture matters more than raw throughput.

Stimulate demand**Nov to Feb**

Use promotions, alternate channels, and tighter inventory planning when both types soften.

Low-demand months require commercial support so the pilot does not confuse seasonality with weak economics.

Management implication: the pilot should be judged against the right seasonal job each month, not against a flat annual operating assumption.

A monthly operating calendar turns seasonality into advantage

READING GUIDE



Service



Protect margin



Stimulate demand



Rebalance / reset

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Commercial posture	Stimulate	Stimulate	Support	Support	Service	Balance	Protect	Protect	Protect	Rebalance	Stimulate	Stimulate
Supply posture	Flex up	Flex up	Flex up	Flex up	Peak flow	Normalize	Tighten mix	Tighten mix	Tighten mix	Reset	Flex down	Flex down
Operating focus	Forecast	Promote	Allocate	Allocate	Service	Audit	Margin	Margin	Margin	Plan	Clear	Review

JAN TO MAY

Service + flow

Prove availability while market volume is naturally strongest.

JUL TO SEP

Margin season

Allocate tightly and protect price in the premium window.

NOV TO FEB

Demand support

Use promotions and lean inventory to protect demand.

Growers need flex contracts, cold-chain discipline, and backup lanes

Build seasonality discipline into contracts and execution rules before the pilot scales.

LEVER

HOW IT SHOULD WORK

CRITICAL OPERATING MOVE

WHY IT MATTERS

Contract tiers

Base + flex volume

Lock a stable base commitment, then layer in seasonal flex volume so supply can expand without year-round overcommitment.

Prevents the pilot from buying peak-season optionality with permanent fixed cost.

Cold-chain readiness

Protect service

Tighten transit monitoring, storage discipline, and receiving checks during the months when flow is highest.

Spoilage and service misses can erase good city economics faster than freight swings do.

Alternate lanes

Reduce disruption

Pre-qualify backup carriers and routes before the peak season so lane shocks do not force reactive reallocations.

The pilot should never have to choose between service and margin because a lane fails.

OPERATING RULE

Contract base volume, protect the cold chain, and pre-book fallback lanes before peak flow starts.

Trade associations should protect off-peak retailer demand

Aim off-peak support at buyers, trade calendars, and food-service pull, not generic consumer messaging alone.

OFF-PEAK BUYER PLAYBOOK

- 1 RETAIL PROMO CALENDAR**
Concentrate retailer ads, features, and display support in Nov-Feb when baseline turns are softest.
- 2 FOOD-SERVICE CHANNELS**
Lean harder on restaurants, prepared foods, and institutional buyers when retail pull weakens.
- 3 TRADE FUNDS + SLOTTING**
Use trade dollars, temporary allowances, and slotting support to win attention in soft months.
- 4 SHELF DISCIPLINE**
Coordinate assortment, facing, and freshness standards so promotions convert to repeat orders.

WHERE TO PUSH HARDER

SOFTTEST MONTHS

Nov to Feb

PRIMARY BUYER

Retail + FS

- Do not mistake seasonal softness for launch failure.
- Push through buyer calendars, not just consumer media.
- Separate repeat demand from promo-driven lift.

EXECUTION RULE

Front-load trade support

Spend trade dollars and retailer attention in Nov-Feb, not when summer demand is already doing the work.

Monthly S&OP should align growers, West Valley, and retailers

One monthly forum should translate market signals into volume, routing, and promotion decisions.

GROWERS

Supply commitment

Provide base volume, flex capacity, and harvest timing updates.

WEST VALLEY

Control tower

Convert price, freight, and demand signals into city allocation decisions.

RETAILERS

Demand signal

Feed promo calendars, replenishment risk, and reorder visibility into the review.



CADENCE

Monthly

Update commitments and allocations every month, not quarterly.

DECISION OUTPUT

Commit + allocate

Reset grower volume, lanes, and promo timing in the same review.

SUCCESS SIGNAL

Margin + service

Higher realized margin should not come with fill-rate or freshness slippage.

Launch in 2026, then scale only what the pilot proves

Treat year one as a live test of economics, service, and demand quality, not a commitment to broad expansion.

ROLLOUT ROADMAP

2026**PILOT**

Launch the five-city test and prove realized margin, fill rate, and spoilage control.

What must be true: city economics hold in live execution.

2027**EXPAND SELECTIVELY**

Add retailer doors only in markets where reorder behavior and service levels remain strong.

What must be true: economics survive broader coverage.

2028**OPTIMIZE SYSTEM**

Tighten grower contracts, allocation logic, and seasonal playbooks around what the pilot confirms.

What must be true: process gains are repeatable.

SCALE ONLY IF

**MARGIN**

Realized margin beats the base case after freight and spoilage.

**SERVICE**

Fill rate holds through peak-season pressure.

**DEMAND**

Retailers reorder because demand is repeatable, not launch noise.

Pilot KPIs should prove margin, service, spoilage, and repeat orders

HEALTH CHECK

GROSS MARGIN

Proof that premium pricing survives live execution.

FILL RATE

Proof that service can scale without breaking.

REORDER RATE

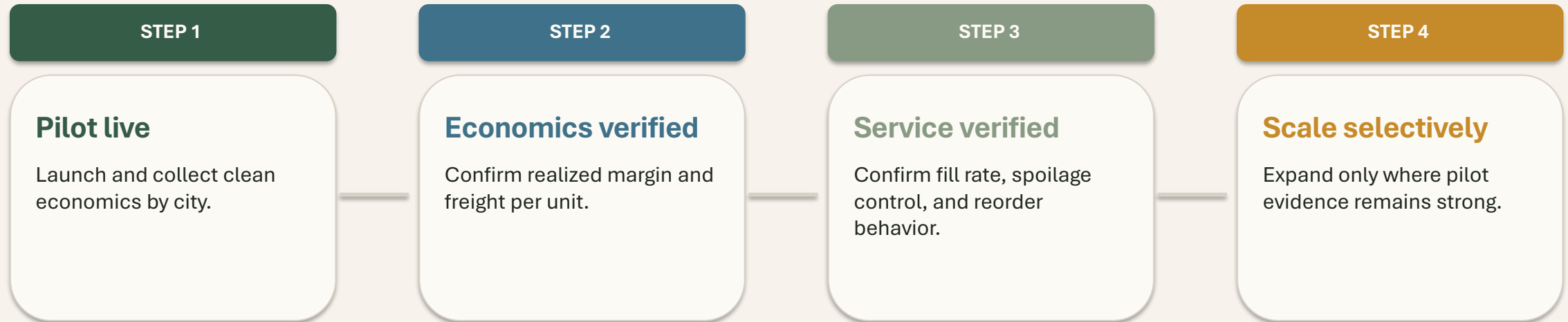
Proof that demand repeats after launch week.

PILOT SCORECARD

KPI	Why it matters	Cadence / owner
Realized gross margin	Confirms whether forecast economics survive live execution.	Weekly / finance
Freight per unit	Shows whether routing and carrier discipline are improving.	Weekly / logistics
Fill rate	Measures retailer execution quality during seasonal peaks.	Weekly / operations
Spoilage rate	Captures cold-chain and freshness failures directly.	Weekly / operations
Reorder rate	Tests whether demand is repeatable beyond launch.	Monthly / commercial
Forecast accuracy	Shows whether monthly S&OP is improving allocation decisions.	Monthly / planning

Scale should follow stage gates, not geography ambition

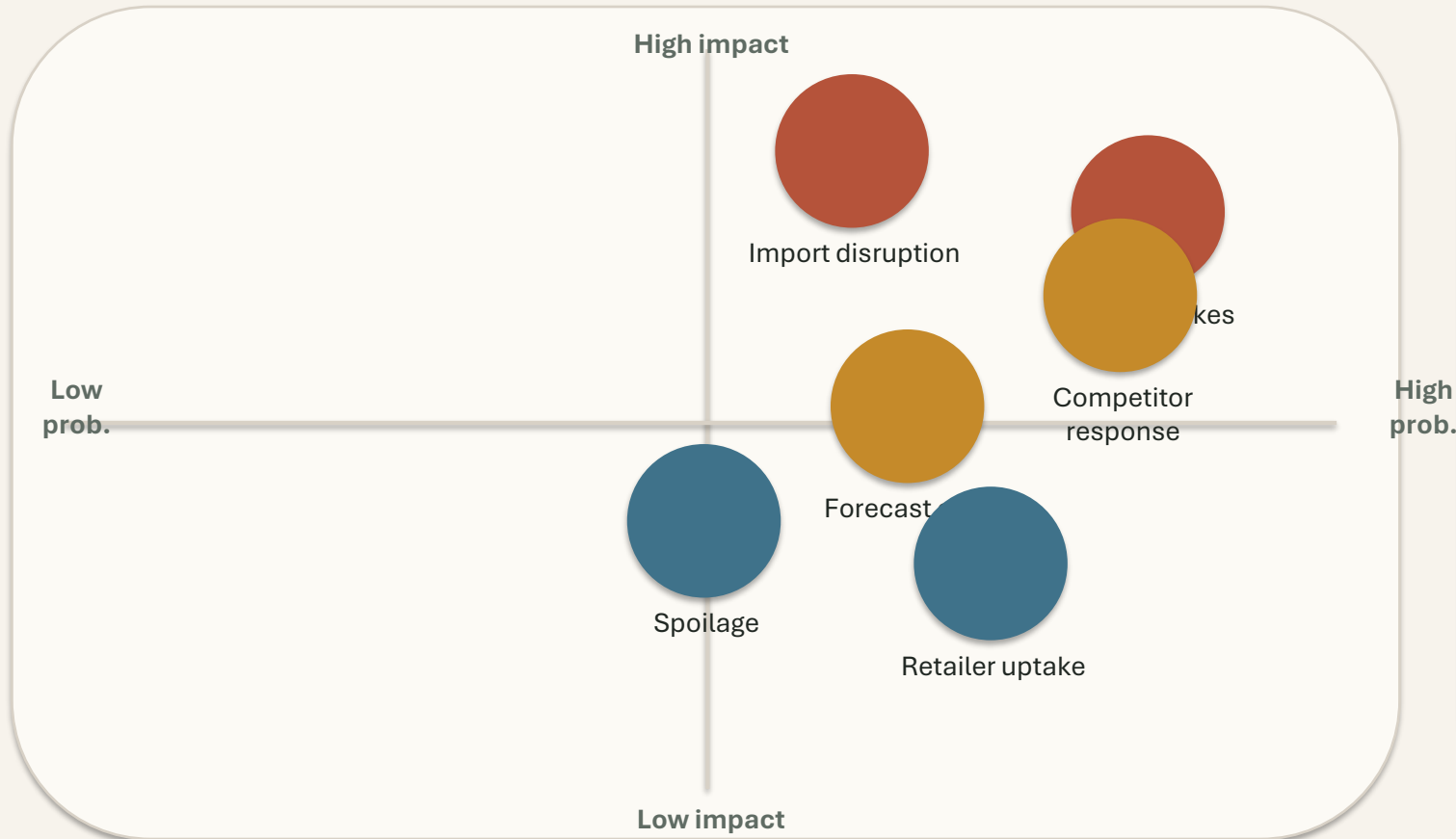
The model can choose where to test, but scale should still depend on live pilot evidence.

**TAKEAWAY**

Use the pilot to validate economics and service before adding geography.

The model is strong enough to choose where to test, not yet strong enough to dictate exact retailer-level allocation.

Top risks are freight, competitors, import disruption, and forecast error



WATCHLIST

FREIGHT SPIKES

Pre-book core lanes and keep alternates ready before peak season.

COMPETITOR RESPONSE

Expect promotions or buyer pressure from Central Valley rivals and hold a firm margin floor.

IMPORT DISRUPTION

Use flexible grower commitments and staggered sourcing windows.

FORECAST ERROR

Re-run allocation monthly using realized pilot demand.

Each major risk already has a countermeasure

Every major threat should already have a response and owner before West Valley scales.

Risk	Trigger	Countermeasure	Owner
Freight spikes	Peak-season lane tightness	Lock core carrier capacity and maintain alternate lanes.	Logistics lead
Import or supply disruption	Supply interruption or customs delay	Use flexible grower contracts and staggered commitments.	Sourcing lead
Spoilage / service failures	Shrink or weak fill-rate trend	Tighten cold-chain controls and receiving discipline.	Operations lead
Forecast error	Demand misses by city	Run monthly S&OP and reallocate by city using realized demand.	Commercial lead
Retailer uptake risk	Weak reorder behavior	Require stage gates before expanding beyond the pilot.	General manager

Operating principle: the pilot should never carry an unmanaged risk; every major threat must already have a trigger, action, and owner before scale decisions are made.

The model is ready for city choice, not exact allocation

What we can defend now

- The cleaned dataset resolves to 40 city markets even though the case text references 41 metro regions.
- The model is strong enough to choose where to test because every city is scored on the same 20,000-unit basis.
- The ranking is resilient: all 40 cities remain profitable even at +50% shipping, so this is an optimization problem.

What still needs more data

- 2025 is partial-year data only and June 2025 retail price is used as the simple proxy for June 2026 retail price.
- The model does not include shrink, retailer-level demand curves, or a fully expanded operating cost stack.
- That means the deck can defend city selection now, but not exact retailer-by-retailer allocation yet.

NEXT DATA TO COLLECT

Pilot reorder behavior, lane-level spoilage, and retailer response would move West Valley from city choice to exact allocation.

Launch the 5-city pilot, then scale only with proof

FINAL RECOMMENDATION

Launch Seattle, Boise, Portland, Spokane, and San Diego first.

The recommendation holds because premium pricing stays stronger than modeled freight swings and seasonality can be managed.

SEATTLE

BOISE

PORTLAND

SPOKANE

SAN DIEGO

AVG PROFIT

\$13.7k

TOP-5 MARGIN

69.7%

FREIGHT RESULT

Same 5 cities

ECONOMICS

\$13.7k avg profit

The top five produce a meaningfully better profit tier than the next options.

RESILIENCE

Same 5 cities

The ranking is unchanged in the base, +50%, and -50% freight cases.

OPERATING DISCIPLINE

Pilot before scale

Scale only where live margin, service, and

Next step: launch, measure live economics, then scale selectively rather than assuming the model is already fully proven.

Appendix: sources, assumptions, and backup exhibits

Primary sources

- case_full.md and case_context.md
- case_comp_data.csv
- outputs/question_1.md through outputs/question_9.md
- q5 to q7 profitability scenario CSVs
- q8 seasonality CSVs and charts

Core assumptions

- 20,000-unit comparable city scorecards
- June 2025 retail price used as June 2026 proxy
- Wholesale price modeled at 40% of retail
- \$0.20 acquisition cost per avocado
- Shipping = \$1,500 fixed + \$50 per 100 miles

Backup exhibits available

- Winner matrix and city trend lines
- Growth/share bridge and elasticity visuals
- Selection matrix, frontier zoom, and freight sensitivity
- Seasonality price and volume charts
- Model caveats and implementation stage gates

Use this appendix to answer questions on data scope, scenario assumptions, seasonality, or why the five-city recommendation stays stable under freight stress.